How Much Must a Developer Invest to Top a Multi-Task Leaderboard? Exact Minimum Targeted Investment under Different Aggregation Rules

Anonymous Author(s)
Affiliation
Address
email

Abstract

A multi-task leaderboard turns per-task scores into one ranking, and a developer can try to move a chosen model to first place by training on data related to some of the tasks. We ask how many score points, over how many tasks, first place requires, and how this depends on the aggregation rule. Investment is the points raised on chosen tasks up to a per-task cap, at a linear, task-symmetric cost; we compute its exact minimum for the mean, geometric, harmonic and trimmed means, median, minimum, Borda, plurality and the Condorcet-winner requirement on HELM MMLU (57 subjects, 79 models) and the Open LLM Leaderboard (6 tasks, 4497 models). Two findings hold on both datasets: the minimum rule is cheaper than the mean in points for every target, most for balanced targets (the cost ratio rises with the coefficient of variation of the target's scores), and rank alone explains most of the variance in cost, the gap to the best rival adding explanatory power only for some rules. Other findings are conditional. For targets ranked 11–30 the median uniform per-task gain needed under the mean is 8.0 points (HELM) and 5.4 points (Open LLM Leaderboard), against a pooled 95th-percentile fine-tuning gain of 25.3 points, whereas the median over all Open LLM Leaderboard targets, 98% of which rank beyond 100, is 30.1; under per-task calibrated caps most targets cannot reach first place, and for HELM this depends on how the gains are transferred (3.2 or 25.3 points); and the order of rules under fixed gains differs between datasets and, on the Open LLM Leaderboard, between data slices. We mark derived identities apart from measurements and report what failed or was not done.

1 Introduction

Multi-task benchmarks are central infrastructure for comparing machine-learning models: a leaderboard has several sub-datasets, each model has a score on each, and an aggregation rule such as the mean or the median produces the ranking. Rankings influence adoption, publication and funding, so a developer has an incentive to push a particular model up. One lever is to fold data related to some sub-datasets into that model's training, which raises its scores there [Dominguez-Olmedo et al., 2024, Zhou et al., 2023]. How much such targeted investment does first place require, and how does the answer depend on the aggregation rule?

We answer this on observed scores, without training any model. A target model may raise its score on any task up to a per-task cap, standing in for what training on related data could buy; all other models are fixed. For each rule we compute the *exact* minimum investment that makes the target strictly first, and report three measures: the total points raised (L_1), the number of tasks touched (K), and the largest single-task raise (L_∞, the cost of a uniform "general improvement"). The caps are the modelling assumption on which every conclusion is conditional, so we vary them: a ceiling, the best

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existing score per task, a fixed gain per task, and a cap calibrated on observed fine-tune-minus-base differences. We also compute the *minimum cap* h^* at which first place becomes feasible.

Cost model. Investment is measured in score points with a linear, task-symmetric cost: a point costs the same on every task and at every level. This is a deliberate simplification, not a claim about what a developer spends; the three measures bracket it (K counts tasks, i.e. the fixed cost of a data campaign; L_∞ the cost of a uniform improvement), and Section 6 states what it leaves out.

Contributions. We separate what follows from the definitions from what we measured, and, among the measurements, what holds on both datasets from what depends on the cap, the sample or the data slice. Each claim is followed by its verdict, including those that contradicted what we expected before running the grids.

- **Derived (Section 4); not empirical.** For the mean, $L_1^* = n\,L_\infty^*$ whenever no cap binds, so targeting saves no points, only concentrates them in fewer tasks; for the minimum and the median we give closed forms, with $K^* \leq (n+1)/2$ for the median. Frontier caps make first place reachable under every monotone rule by construction. The solvers agree with the formulas (Section 5.2).
- **Measured, on both datasets.** Contrary to our expectation, the minimum is cheaper than the mean in points for every target under the linear cost (for 97% and 88% of targets under a quadratic cost; Appendix D), and relatively cheapest for balanced targets (the cost ratio rises with the coefficient of variation of the target's scores, Spearman $\rho = +0.92$ on HELM, +0.84 on the Open LLM Leaderboard; other dispersion statistics are not stable). The median is not cheaper than the minimum in tasks touched at any task count we tried, again contradicting our expectation (Section 5.2). Cost grows with rank, and rank alone already explains most of the variance; the gap to the best rival adds explanatory power for the geometric and harmonic means and the median on both datasets and for Borda on HELM, but not for the minimum, the Condorcet requirement, Borda on the Open LLM Leaderboard or plurality on HELM (Section 5.3); for the first three the gap is close to the cost by construction, so this is a weak test.
- **Measured, but conditional on the cap, the sample or the slice.** Whether first place is reachable depends mainly on the cap. Under a fixed gain per task the order of rules differs between the datasets and changes with the margin by which first place must be won (appendix), and on the Open LLM Leaderboard the order of the median and the minimum rule's cost in points also changes with the data slice (appendix; slice medians use different target sets). The median h^* is reported by rank bin: for ranks 11–30 it is 8.0 (HELM) and 5.4 (Open LLM Leaderboard) points under the mean, against a pooled 95th-percentile fine-tuning gain of 25.3 points, while the all-target median is dominated by the 98% of Open LLM Leaderboard targets ranked beyond 100. Under per-task calibrated caps most targets cannot reach first place, and for HELM this depends on the transfer of the fine-tuning gains (3.2 points from one task, 25.3 pooled; Section 5.4). We present this as a sensitivity, not as a prediction.
- **Ordinal rules differ in kind.** On HELM (ranks 11–30, ceiling) plurality is cheapest in both measures, while Borda and the Condorcet requirement need many tasks; contrary to our expectation, the Condorcet requirement is cheaper than the mean in points (Section 5.5). On the Open LLM Leaderboard, for all targets of ranks 2–100, plurality is again the hardest rule to make feasible under a fixed gain (14% of targets at cap $+10$).

We discuss the assumptions behind these results, what breaks them, and what we did not do in Section 6. Related work is in Section 2.

2 Related Work

Multi-task leaderboards aggregate per-task scores into one ranking. We study two: the HELM MMLU subject tables [Liang et al., 2022, Hendrycks et al., 2021] and the Open LLM Leaderboard [Hugging Face Open LLM Leaderboard, 2025], whose six task families include BBH [Suzgun et al., 2023] and are scored with the evaluation harness of Gao et al. [2023] (Section 5.1).

Fragility of aggregation. Dehghani et al. [2021] show, on SuperGLUE, VTAB, Long Range Arena and Atari, that which tasks are bundled and how they are combined (mean or median) changes which method comes first. Zhang and Hardt [2024] view a benchmark as an election with tasks as voters, separate cardinal from ordinal aggregation, and measure a trade-off between a benchmark's diversity and its sensitivity to irrelevant changes. Lanctot et al. [2023] also treat tasks as voters and propose social-choice aggregators with robustness properties such as clone-proofness; Singh et al. [2025] document how private testing and uneven data access distort the Chatbot Arena ranking. These works vary the *benchmark*, its task or model set or its aggregator. We hold the benchmark fixed and vary the *developer's action*: how many score points, over how many tasks, must be added to one model's scores to make it first under a given rule.

Training on the test task. Using evaluation data, or knowledge of the evaluation tasks, at training time raises scores and changes model comparisons [Zhou et al., 2023, Yang et al., 2023, Dominguez-Olmedo et al., 2024]. We take such gains as the available lever and train no model: the per-task *cap* on the gain (Section 3) stands in for what training can buy, and we vary it, including a cap calibrated on observed fine-tune-minus-base differences.

Strategic post-training. The nearest formal treatment of developer incentives is Chen et al. [2026], who model a single benchmark as a Stackelberg game between a designer and competing developers, each choosing an effort that yields a post-effort score, prove that current benchmarks admit no Nash equilibrium among developers, and show that tune-before-test yields an equilibrium ranking by latent quality. Their model has a scalar score per model, so the aggregation of several tasks into one ranking does not arise there, and it studies equilibrium behaviour. Ours is a single developer facing frozen rivals, solved exactly on observed scores, so the two address different quantities.

Bribery and manipulation of scoring rules. The closest analogue in computational social choice [Brandt et al., 2016] is manipulation and bribery: changing the inputs of an election at minimum cost until a preferred candidate wins [Bartholdi et al., 1989, Faliszewski et al., 2009, Faliszewski and Procaccia, 2010], in particular shift and swap bribery, where a bribed voter moves the candidate up its ranking and the cost is the number of positions [Elkind et al., 2009, 2020]. Those models act on ordinal votes of a population of voters and the question is mostly computational complexity. Ours differs in three respects. The inputs are cardinal scores of a few tasks, the lever is a raise of one model's scores bounded by a per-task cap with cost in score points, and the questions are empirical: how the minimum cost varies across rules, targets and caps on observed leaderboards. The individual solvers are elementary (water-filling for power means, thresholding for the median and the minimum, mixed-integer programs for the ordinal rules and the trimmed mean); what we add is the rule-by-rule comparison of exact minima, the minimum feasible cap h^*, and the measurements on two leaderboards. We do not claim a new complexity result.

Choice of rules. Real leaderboards mostly use the mean: HELM reports a mean win rate over scenarios, the Open LLM Leaderboard the mean of its normalised task scores. We include the geometric and harmonic means and the median as the other common cardinal summaries, the minimum as the limiting case of a worst-task rule, the trimmed mean as a compromise, and Borda, plurality and the Condorcet requirement because the social-choice treatments of benchmarks [Zhang and Hardt, 2024, Lanctot et al., 2023] study ordinal as well as cardinal aggregation of tasks. Plurality and Condorcet are not used by the leaderboards we study; they serve to show how the cost depends on whether scores or ranks are aggregated.

3 Problem Formulation

Leaderboard. A leaderboard is a matrix $X \in [0, 100]^{M \times n}$: M models, n tasks, X_{jt} the score of model j on task t. An aggregation rule maps X to a ranking of the models. *Score rules* assign each model a value $f(X_j)$ from its own row: mean, geometric and harmonic mean, median, minimum, trimmed mean. *Rank rules* use comparisons between models: Borda (the sum over tasks of the number of models beaten, ties counting one half), plurality (the number of tasks on which the model is best), and the Condorcet requirement (the target beats every other model on a majority of tasks).

Targeted investment. A developer holds a target model i with scores $x = X_i$. Training on data related to task t raises its score to any $y_t \in [x_t, c_t]$; all other models are fixed. The *cap* vector c is the modelling assumption on what training can buy. We study caps of the form ceiling ($c_t = 100$), frontier (c_t is the best score of any other model on t plus 0 or 2 points), additive ($c_t = x_t + h$, $h \in \{5, 10, 20\}$) and empirical (the 95th or 99th percentile of per-task fine-tune-minus-base differences measured on the Open LLM Leaderboard). Target i is *first* under rule f if its aggregate exceeds every other model's by at least $\varepsilon = 10^{-3}$ points.

Investment measures. For a raise y that makes the target first we report three measures: the total score gain $L_1 = \sum_t (y_t - x_t)$, the number of tasks touched $K = |\{t : y_t > x_t\}|$, and the largest single-task jump $L_\infty = \max_t (y_t - x_t)$, each minimised separately. If no $y \le c$ makes the target first, the target is *infeasible* under that rule and cap, and we report infeasible counts rather than dropping those targets.

Minimum cap. To see how feasibility depends on the cap rather than only at a few scenarios, we also compute for each rule and target h^*, the smallest uniform per-task gain cap h (with $c_t = \min(x_t + h, 100)$) under which first place is feasible.

4 Exact Minimum-Investment Solvers

For every rule and cap we compute the exact optimum, not a heuristic. For a score rule f let $\tau = \max_{j \neq i} f(X_j)$ and $b = \tau + \varepsilon$, so that first place means $f(y) \ge b$; let $H_t = c_t - x_t$ be the headroom on task t. L_1, K and L_∞ are three separate optimisations, so the K-minimal and the L_1-minimal raise generally differ. All rules are monotone in the target's own scores, hence first place is reachable under a cap if and only if the target raised to the cap c is first; every solver uses this as its feasibility test.

Proposition 1 (Mean)**.** *If $\sum_t H_t \ge nb - \sum_t x_t$ then $L_1^* = \max(0, nb - \sum_t x_t)$ and K^* is the smallest k such that the k largest headrooms sum to at least L_1^*; otherwise first place is infeasible. If no cap binds, $L_\infty^* = b - \bar{x}$ and $L_1^* = n\, L_\infty^*$.*

Under the mean, spreading the investment costs exactly as many points as concentrating it: targeting buys a smaller K, not fewer points.

Proposition 2 (Minimum)**.** *First place is feasible iff $c_t \ge b$ for all t. Then $L_1^* = \sum_t (b - x_t)_+$, $K^* = |\{t : x_t < b\}|$, $L_\infty^* = \max_t (b - x_t)_+$, so $L_1^* \le n L_\infty^*$.*

Proposition 3 (Median, n odd)**.** *With $m = (n+1)/2$, first place is feasible iff at least m tasks have $c_t \ge b$. Then $K^* = \max\big(0, m - |\{t : x_t \ge b\}|\big) \le m$ and L_1^* is the sum of the m smallest shortfalls $(b - x_t)_+$ among tasks with $c_t \ge b$.*

Proofs are in Appendix A. For even n (Open LLM Leaderboard, $n = 6$) the median is the mean of the two middle scores and the solver enumerates the pair of threshold levels. Power means with exponent $p < 1$ (geometric: $p \to 0$; harmonic: $p = -1$, both on $y + 1$) are $\frac{1}{n} \sum_t g(y_t) \ge g(b)$ with g increasing and concave; L_1^* is then a water-filling problem (bisection on the common level $y_t + 1$, clipped to $[x_t, c_t]$) and K^* is greedy in $g(c_t) - g(x_t)$. The t-trimmed mean has no closed form and is solved as a mixed-integer program (MILP).

Rank rules. A raise of the target on task t to a level just above a rival's score passes every model at or below that score, so only the levels $\{s_{kt}\}$ (tie, half credit) and $\{s_{kt} + \varepsilon\}$ (strict pass) matter, at most one per task. Borda, plurality and the Condorcet-winner requirement are then exact 0/1 programs, solved with HiGHS (SciPy) with lazily generated rival constraints. The Condorcet winner (the target wins strictly more than $n/2$ tasks against every rival) is sufficient for first place under any Condorcet-consistent rule, so its minimum investment is an *upper bound* for that family; the reported "rule rank" for it is a Copeland proxy.

Verification. Propositions 1–3 are proved and also compared with the solvers on all 79 HELM targets (Section 5.2). The trimmed-mean and rank-rule MILPs are exact by construction but come with no separate proof; they are checked against independent formulations (an assignment formulation of the trimmed mean, a grid search for the even median) and against brute-force enumeration on small random instances.

5 Experiments

5.1 Data and setup

We use two frozen public snapshots; no model is trained or evaluated by us. **HELM MMLU** (release v1.13.0, 2025-01-10): the per-subject exact-match tables of 57 MMLU subjects, with all $M = 79$ models present on every subject ($n = 57$). **Open LLM Leaderboard** (`contents`, commit `9c09a7ca`, last updated 2025-03-20): $M = 4497$ models after dropping 79 duplicated names (keeping the best-scoring variant) and $n = 6$ task families (IFEval, BBH, MATH Lvl 5, GPQA, MuSR, MMLU-Pro) on the leaderboard's normalised 0–100 scale; one flagged model is removed, and 708 of the models are merges (kept in the main analysis, dropped in a robustness slice). The two datasets span two regimes: few tasks and many models, and many correlated tasks and few models. Neither source declares a data licence (the HELM code is Apache-2.0); we use public aggregate scores only, redistribute only derived tables and code, and give versions, checksums and our licence judgement in the reproduction notes. The analysis runs on CPU only and trains no model.

Every model whose rank under the mean is at least 2 is a target (HELM: 78 targets; Open LLM Leaderboard: 4495 for the score rules). Ordinal rules on the Open LLM Leaderboard are run on a rank-stratified sample of targets (fixed seed, drawn once) because their mixed-integer programs are costly; for ranks 2–100 they were also run on every target (Appendix D). Each target and (rule, cap) pair is solved exactly with the solvers of Section 4, margin $\varepsilon = 10^{-3}$. Targets are the whole population of each snapshot, so intervals over targets are descriptive: they are 95% percentile bootstrap intervals over the targets in a bin, and Wilson intervals only for the shares in Figure 3. They cover the sampling of targets within one snapshot and nothing else: not the choice of snapshot, of task set or of submitted models, not evaluation noise (Section 5.6), and not the dependence between rows, since many models are fine-tunes or merges of one another. They should be read as a measure of spread, not of the precision of a population parameter.

Empirical gains. The empirical caps use the 540 (fine-tune, base) pairs of the Open LLM Leaderboard for which the model's `Base Model` field names another unflagged model of the leaderboard (the field is user-entered metadata and we did not audit it); the gain on a task is the fine-tune's score minus the base's. The pairs mix instruction and chat tunes, merges and domain fine-tunes and are not outcomes of targeted training; they are a proxy with unknown selection. A restricted set without chat-typed models, merges and instruction-named tunes (239 pairs) gives a pooled 95% quantile of 19.0 instead of 25.3 points and nearly the same infeasible shares (Appendix D).

5.2 Closed forms and rule structure

Solvers match the closed forms. On the 237 HELM rows (mean, minimum, median; ceiling cap) the solvers and Propositions 1–3 agree to $|\Delta L_1| \le 2.3 \times 10^{-13}$, $|\Delta K| \le 0$ and $|\Delta L_\infty| \le 9.5 \times 10^{-8}$ (bisection tolerance; Table 2, appendix). Under the ceiling the equality $L_1^* = nL_\infty^*$ of Proposition 1 applies to only 4 of 79 HELM targets, because a uniform raise clips at 100 for most; in general $L_1^* \le nL_\infty^*$, with equality when no cap binds.

Minimum versus mean. For every target with $L_1 > 0$ (100% of the 78 HELM targets and 100% of the 4495 Open LLM Leaderboard targets) the minimum rule needs fewer points than the mean, so the expectation that the minimum is dearer for balanced targets is contradicted. The median ratio $L_1^{\min}/L_1^{\text{mean}}$ is 0.15 on HELM (95% interval 0.12–0.18) and 0.25 on the Open LLM Leaderboard (0.25–0.25). With the dispersion of a target's scores measured by the coefficient of variation (CV) of its task scores, the ratio *rises* with CV on both datasets: the median ratio in the low, middle and high CV terciles is 0.08, 0.15 and 0.23 on HELM (Spearman $\rho = +0.92$) and 0.20, 0.24 and 0.35 on the Open LLM Leaderboard ($\rho = +0.84$): the minimum saves most for balanced targets. This is specific to the statistic: with the spread measured as mean minus minimum, ρ is +0.27 on HELM and -0.51 on the Open LLM Leaderboard, so the dependence on dispersion is not stable across statistics and datasets. Cheaper in points does not mean more often first: under additive gain caps the minimum is the least reachable rule on HELM (Section 5.4), so cost and feasibility must be read together.

Task-subset scaling. We draw random task subsets of $n = 5, \ldots, 57$ from HELM (50 draws, cap $+10$, two targets of rank 2 and 5, so this is a small study; Figure 5). In K/n at $n = 57$ the mean

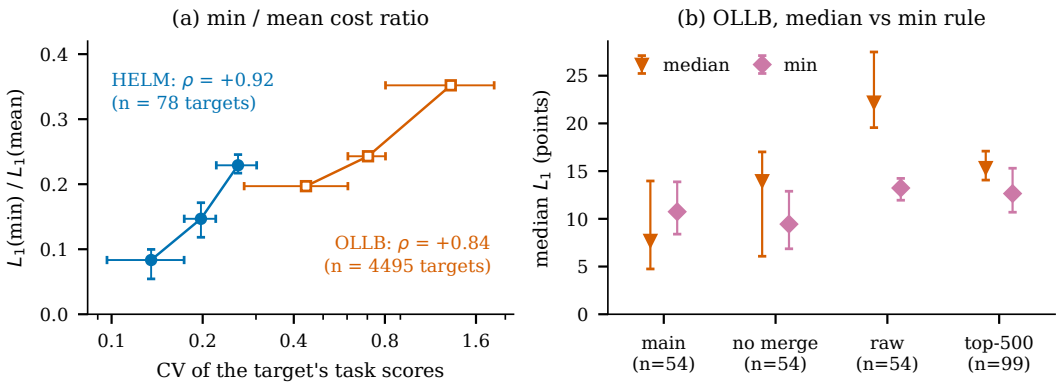


Figure 1: (a) Cost of the minimum rule relative to the mean, $L_1^{\min}/L_1^{\text{mean}}$ at cap 100: median per tercile of the coefficient of variation (CV) of the target's scores; horizontal bar the tercile's CV range, vertical bar the 95% bootstrap interval over targets; HELM filled (78 targets), Open LLM Leaderboard open (4495); Spearman ρ over all targets is printed. (b) Open LLM Leaderboard, targets ranked 2–100: median L_1 of the median rule and of the minimum rule on four data slices (main sample, no merged models, raw scores, top 500) with 95% bootstrap intervals and n targets; compare the order within a slice, not levels across slices.

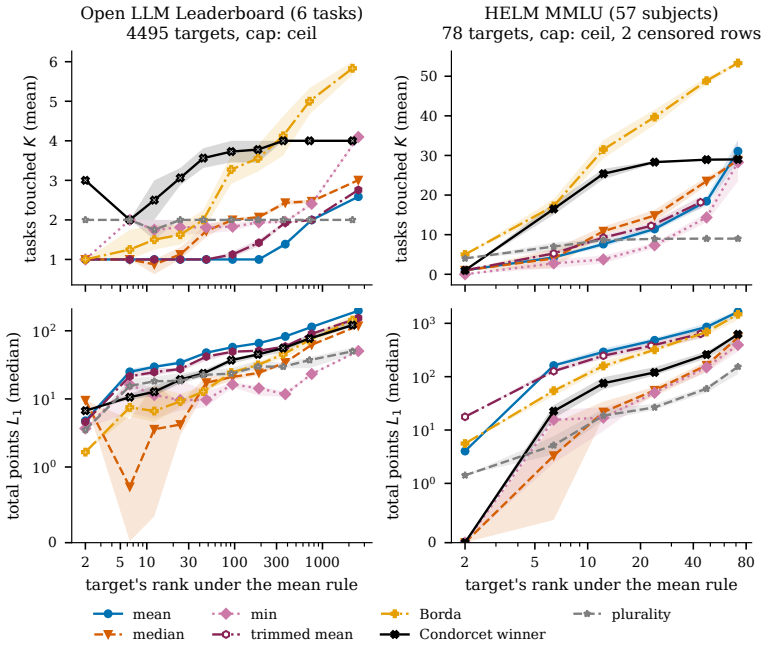


Figure 2: Minimum investment to reach first place versus current rank under the mean (cap 100). Top: tasks touched K (mean per rank bin); bottom: points L_1 (median per bin, symlog axis). Bands: 95% bootstrap intervals over the targets in a bin. Left: Open LLM Leaderboard; right: HELM MMLU. K-minimal and L_1-minimal solutions are separate optima.

needs 0.15, Borda 0.21, plurality 0.06, the median 0.04 and the minimum 0.03. The median is above the minimum at every n tested, so the expectation that the median is cheaper than the minimum in tasks touched was not borne out; at $n = 5$ the minimum is infeasible for 0.18 of the draws of the rank-5 target.

5.3 Cost versus current rank

Figure 2 shows that on both datasets cost grows with rank, with rule-specific slopes. For the mean the dependence is an identity: L_1^* is linear in the gap to the best rival (Proposition 1), so $R^2 = 1$ against the gap is not evidence. For the geometric and harmonic means and the median the cost is also close to a function of the gap in rule units by construction (the rule value must rise by the gap), so the informative quantity is what the gap adds to rank. We regress, for every rule but the mean, $\log L_1$ on $\log$(gap in rule units) and on $\log$(rank) (ceiling cap, targets with positive gap and cost) and report the difference of the two R^2 with a bootstrap interval over targets (Table 3, appendix). The gap beats rank, with an interval for the R^2 difference that excludes zero, for the geometric mean (R^2 1.00 vs. 0.92 on HELM, 0.97 vs. 0.94 on the Open LLM Leaderboard), the harmonic mean and the median on both datasets, and for Borda on HELM (0.96 vs. 0.91); on the Open LLM Leaderboard the margins are small (0.02–0.03). It is *not* supported for the minimum (difference -0.014 [-0.037, +0.005] on HELM, -0.005 [-0.007, -0.003] on the Open LLM Leaderboard), the Condorcet requirement (+0.057 [-0.003, +0.077]; -0.008 [-0.022, +0.006]), Borda on the Open LLM Leaderboard or plurality on HELM; Table 3 gives all values. Plurality on the Open LLM Leaderboard cannot be tested because rank ties leave too few distinct costs, and the ordinal rules there use a rank-stratified sample, so their intervals are wide. Rank alone is already a strong predictor everywhere ($R^2 \geq 0.72$, and ≥ 0.84 for all score rules), so the gap adds explanatory power only where stated above.

5.4 Caps decide feasibility

Frontier caps make first always reachable. Under the ceiling, front+0 and front+2, first place is feasible for every target under every rule on both datasets. This is mechanical: a target raised to the best rival score on every task weakly dominates every rival everywhere, so it is first under any monotone rule. The frontier caps therefore cannot separate rules in feasibility; they only change cost.

Fixed per-task gains separate rules, differently on each dataset. With cap $+10$ the share of HELM targets that can reach first ranges from 73% (plurality) through 51% (median), 37% (mean) and 32% (Borda) to 31% (geometric), 26% (harmonic) and 18% (minimum; Condorcet 49%). On the Open LLM Leaderboard (all 4495 targets, score rules) it is 3% (mean), 5% (median), 3% (geometric), 6% (harmonic) and 10% (minimum): the minimum is the *least* reachable rule on HELM and the *most* reachable of these five on the Open LLM Leaderboard. The ordinal rules there were run on all 99 targets of ranks 2–100 only, where the shares at cap $+10$ are plurality 14%, Condorcet 54%, Borda 93% against 100% for the mean, 76% for the median and 69% for the minimum; these shares are for the top of the board and must not be compared with the all-target figures. Their minimum uniform gain cap h^* has median 4.2 (Borda), 8.3 (Condorcet) and 10.7 (plurality) points for ranks 11–30, against 5.4 for the mean, and at most 19.7 points in ranks 2–100, below the pooled empirical quantile (25.3).

Minimum cap h^* and the empirical caps. The minimum uniform gain cap h^* removes the choice of a few scenarios. It depends strongly on the target's rank, so Table 1 reports it by rank bin. Under the mean, the median h^* is 8.0 points on HELM and 5.4 points on the Open LLM Leaderboard for ranks 11–30, and 19.9 and 8.8 for ranks 31–100; the 4396 Open LLM Leaderboard targets ranked beyond 100 (98% of all) need 30.3. The all-target medians, 13.2 points on HELM (95% interval 10.4–16.5) and 30.1 on the Open LLM Leaderboard (29.7–30.4), are dominated by low-ranked targets and should not be read as the investment of a contender for first place. Across rules the all-target medians run on HELM from 6.2 (plurality) to 22.3 (minimum), with the mean at 13.2, and on the Open LLM Leaderboard from 17.9 (minimum) to 36.2 (median), with the mean at 30.1: the order of the minimum and the median reverses between datasets, and on the Open LLM Leaderboard the order of their L_1 also depends on the data slice (Appendix C). We compare with observed fine-tuning gains: the pooled 95th percentile of 540 fine-tune-minus-base differences on the Open LLM Leaderboard is 25.3 points, and the per-task 95th percentiles range from 3.2 to 46.0. Under the pooled value the share of targets whose h^* exceeds it is 13% (mean, HELM) and 69% (mean, Open LLM Leaderboard); for the minimum it is 38% and 0%. Under the *per-task* 95th-percentile caps, which are far more restrictive, first place is unreachable for 91% of HELM targets under the mean (79% under plurality) and for 93% of Open LLM Leaderboard targets under the mean (99% under the minimum). On HELM these caps are transferred from the MMLU-Pro quantile (3.2 points, against 25.3 for the pooled scalar) because HELM has no base/fine-tune pairs; with the pooled scalar instead the feasible

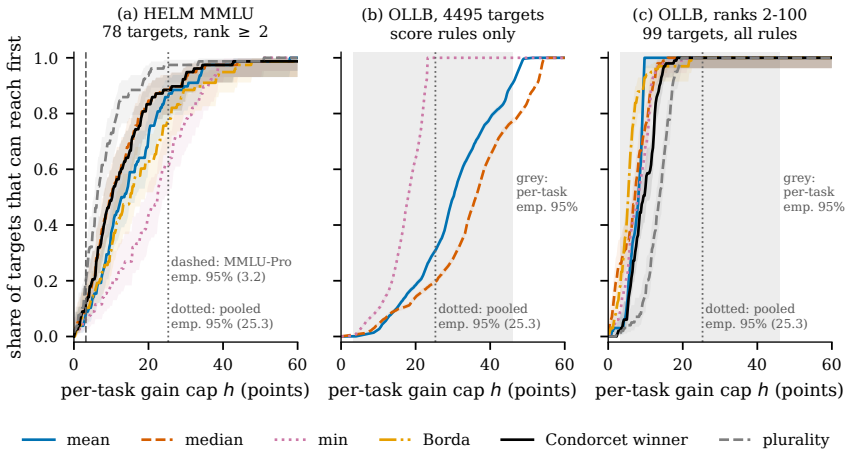


Figure 3: Share of targets with $h^* \leq h$ (95% Wilson intervals). (a) HELM, 78 targets, all rules. (b) Open LLM Leaderboard, all 4495 targets, score rules only. (c) Open LLM Leaderboard, all 99 targets of ranks 2–100, all six rules on the same targets (ordinal rules were solved for these only). Dotted: pooled empirical 95th percentile of fine-tuning gains (25.3 points); grey: its range over the six tasks (3.2–46.0); dashed (HELM): the MMLU-Pro value (3.2).

Table 1: Minimum uniform gain cap h^* (points) for first place by rank bin: median with 95% bootstrap interval over targets (none for fewer than 3 targets) for the mean, median and minimum rules, and the share of targets whose h^* exceeds the pooled 95th percentile of fine-tuning gains (25.3) and, on HELM, the MMLU-Pro value that the HELM empirical caps use (3.2). Infeasible counts as exceeding. 98% of the Open LLM Leaderboard targets rank beyond 100 and dominate its all-target median (30.1); rank 2 is one target. Score rules only.

dataset	rank bin	targets	median h^* [95% CI] (points)			share with $h^* >$	
			mean	median	min	25.3 (pooled)	3.2 (MMLU-Pro)
HELM	2	1	0.1	0.0	0.0	0.00	0.00
	3–10	8	2.9 [0.8, 4.0]	1.5 [0.6, 2.8]	3.9 [1.8, 12.6]	0.00	0.25
	11–30	20	8.0 [6.9, 9.0]	5.8 [4.9, 6.5]	14.6 [9.3, 19.6]	0.00	1.00
	31–100	49	19.9 [16.3, 21.8]	14.6 [12.5, 17.2]	26.6 [23.4, 29.0]	0.20	1.00
OLLB	2	1	0.8	4.8	3.7	0.00	–
	3–10	8	4.2 [1.3, 4.7]	0.4 [0.1, 4.2]	8.1 [3.1, 10.8]	0.00	–
	11–30	20	5.4 [5.3, 5.7]	2.1 [1.5, 4.4]	5.9 [5.2, 9.8]	0.00	–
	31–100	70	8.8 [8.5, 9.0]	8.2 [7.3, 9.5]	8.3 [6.7, 9.6]	0.00	–
	>100	4396	30.3 [30.0, 30.6]	36.5 [36.1, 36.7]	18.1 [17.9, 18.3]	0.71	–

share is 62–97%, so the HELM conclusion depends strongly on this transfer. Under the MMLU-Pro value the share of HELM targets whose h^* exceeds the cap is 79–95% across rules, against 13% (mean) under the pooled scalar.

5.5 Ordinal rules

Table 5 (appendix) gives the ceiling, cap $+10$ and empirical-95% blocks. On HELM, rank bin 11–30, ceiling cap, median K and L_1 are: plurality 9 tasks and 25 points; mean 10 and 435; Condorcet 28 and 107; Borda 38 and 286. Borda and the Condorcet requirement therefore need many tasks, but, contrary to our expectation, the Condorcet requirement is *cheaper* than the mean in points; plurality is cheapest in both. Because the Condorcet requirement is sufficient for first under any Condorcet-consistent rule, these numbers are an upper bound for Copeland, Schulze or Kemeny. On the Open LLM Leaderboard, over all 99 targets of ranks 2–100 (ceiling), Borda, plurality and the Condorcet requirement touch a median of 2, 2 and 3 tasks, but under cap $+10$ plurality is the hardest rule to make feasible (14% of targets, against 93% for Borda and 54% for the Condorcet requirement; Appendix D).

5.6 Robustness

Each draw needs an exact solve, so the noise and task-bootstrap runs use rank-stratified samples of 33 HELM and 29 Open LLM Leaderboard targets (100 and 100 noise draws, 100 bootstrap draws; score rules only on the latter). Under evaluation noise (binomial standard errors from item counts, cap $+10$) the ratio of L_1 to its observed value (median over targets, mean over draws) is 1.04 for the mean and 2.18 for the minimum on HELM, 1.03 and 1.58 on the Open LLM Leaderboard; the minimum rule can reach first place in only 24% of noisy HELM target-draws, against 39% of the targets on the observed scores. A target that is first by exactly $\varepsilon = 10^{-3}$ stays first with probability only about 0.22 (mean), 0.30 (median), 0.24 (minimum) on HELM (ranks 11–30), so ε is not a robust margin. Margins of 10^{-6} and 0.1 change the median L_1 by at most a few per cent, but rule-specific margins halve the minimum's advantage over the mean and collapse its feasibility under fixed gains (Appendix D); data slices change the order of the median and the minimum rule's cost (Appendix C).

6 Discussion and Limitations

Modelling assumptions. The analysis assumes (A1) a target's score on a task can be raised independently of its other scores, (A2) rivals do not react, (A3) training on one task neither helps nor hurts other tasks (no spillover), and (A4) scores lie in $[0, 100]$ and gains are limited by the cap, and (A5) investment costs one unit per score point on every task and at every level (linear, task-symmetric cost). Each can fail: training on one task can move others; rivals can respond [Chen et al., 2026]; ceiling effects make late points dearer: under a quadratic cost the minimum is still cheaper than the mean for 97% (HELM) and 88% (Open LLM Leaderboard) of targets (Appendix D), but we did not vary the price across tasks; and our model does not separate legitimate training from contamination [Zhou et al., 2023, Yang et al., 2023]. The caps are assumptions, not predictions.

Scope. The targets are all models of two frozen snapshots, many of which are fine-tunes or merges of one another, so rows are dependent and intervals over targets are descriptive. The Open LLM Leaderboard has $n = 6$ aggregate tasks and thousands of models; HELM MMLU has 57 subjects of one benchmark, which are strongly correlated, and 79 models. Results do not extrapolate to leaderboards with other sizes or rules. The Condorcet requirement is sufficient for first place under any Condorcet-consistent rule, so its cost is an upper bound for that family; Kemeny and Schulze were not run.

Unfinished, failed and censored analyses. (i) The Open LLM Leaderboard sub-tasks (of BBH and MMLU-Pro) were not analysed; HELM's 57 subjects stand in for many-task behaviour. (ii) On the Open LLM Leaderboard the ordinal rules cover every target of ranks 2–100 and, in the main tables and slices, a rank-stratified sample of 101–102 of the 4495 targets; nothing is said about them beyond rank 100, and the minimum cap h^* of these rules is computed for ranks 2–100 only. The trimmed mean drops $t = 1$ of 6 scores at each end on the Open LLM Leaderboard (all targets) and $t = 5$ of 57 on HELM (36 targets), so its two lines are not comparable; 4 of its 144 HELM solves are censored by the time limit. A censored solve is reported as censored, never as infeasible, and one Borda h^* row, for a target of rank 4496 in the stratified sample, is censored at the solver limit (it counts neither as feasible nor as infeasible); no other censored row occurs in the main grids or the h^* files. (iii) The noise and task-bootstrap experiments use rank-stratified samples of 33 (HELM) and 29 (Open LLM Leaderboard, score rules only) targets; they are not population estimates. (iv) Earlier runs with out-of-memory failures, a wrong trimmed-mean formulation or time-outs counted as infeasible were discarded and re-run. (v) The empirical-gain caps rest on 540 fine-tune/base pairs from one leaderboard, not on *targeted* training; on HELM they are transferred from one task.

Ethics and societal impact. All data are public aggregate scores; no model is trained and no human subjects or private data are involved. The study can inform benchmark designers about which rules are cheap to move, and can be read as a cost estimate for gaming a leaderboard; no developers are identified.

Use of LLM-based agents. Planning, code, figures and this text were produced by LLM-based agents; the solvers use no language model, and every number in the text is generated from the result files by `paper/make_numbers.py`.

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A Proofs of Propositions 1–3

Throughout, $b = \tau + \varepsilon$ and $H_t = c_t - x_t$.

Proposition 1 (mean). The mean is linear: $\bar{y} \geq b$ iff $\sum_t (y_t - x_t) \geq nb - \sum_t x_t$, and any $y \in [x, c]$ with that total raise is feasible, which requires $\sum_t H_t \geq nb - \sum_t x_t$ and gives $L_1^* = \max(0, nb - \sum_t x_t)$. K is minimised by filling the largest headrooms first. If no cap binds, a uniform raise d gives mean $\bar{x} + d$, so $L_\infty^* = b - \bar{x}$ and $nL_\infty^* = nb - \sum_t x_t = L_1^*$. In general a clipped uniform raise satisfies the constraint with total at most nL_∞, hence $L_1^* \leq nL_\infty^*$. □

Proposition 2 (minimum). $\min_t y_t \geq b$ iff $y_t \geq b$ for all t; raising task t to exactly $\max(x_t, b)$ is cheapest and possible iff $c_t \geq b$ whenever $x_t < b$; since $c_t \geq x_t$, this is the same as $c_t \geq b$ for all t. Tasks with $x_t \geq b$ are untouched, and the costs follow. □

Proposition 3 (median, n odd). With $m = (n+1)/2$, $\text{median}(y) \geq b$ iff at least m coordinates are $\geq b$. A task already at $x_t \geq b$ counts at no cost; every other task can be lifted to exactly b at cost $b - x_t$ if $c_t \geq b$. The cheapest way to have m coordinates $\geq b$ lifts the tasks with the smallest shortfalls $(b - x_t)_+$ among those with $c_t \geq b$, which is L_1^*; $K^* = \max(0, m - |\{t : x_t \geq b\}|) \leq m$. Feasibility needs at least m tasks with $c_t \geq b$. □

Monotonicity. Every rule here is nondecreasing in the target's own scores, so the set of feasible raises within the cap box is an up-set, and first place is reachable under cap c iff the target raised to c is first.

B Additional results

Table 2: Closed forms re-derived independently of the solvers against solver output on HELM (79 models, cap 100): maximum absolute difference per rule; the number in brackets is the number of targets to which the uniform-raise form of L_∞ applies.

rule and closed form	targets	$\max \lvert\Delta L_1\rvert$	$\max \lvert\Delta K\rvert$	$\max \lvert\Delta L_\infty\rvert$ (n)
mean ($L_1^* = n(\tau + \varepsilon) - \sum x$)	79	2.3e-13	0	7.6e-08 (4)
min ($L_1^* = \sum(b - x)_+$)	79	2.3e-13	0	9.2e-08 (79)
median ($K^* = \frac{n+1}{2} - \#\{x \geq b\}$)	79	1.1e-13	0	9.4e-08 (79)

Table 3: R^2 of $\log L_1$ on $\log$ rank and on $\log$ gap (ceiling cap; targets with positive gap and cost), with 95% bootstrap intervals over targets (descriptive). The last column is the difference of the two R^2 with its interval; for the mean the gap fit is an identity. Ordinal rules on the Open LLM Leaderboard use the rank-stratified sample; plurality there is untestable (rank ties).

dataset	rule	targets	R^2 on log rank	R^2 on log gap	difference (gap − rank)
HELM	mean	78	0.899 [0.872, 0.953]	1.000 [1.000, 1.000]	0.10 [0.05, 0.13]
HELM	geometric mean	77	0.920 [0.902, 0.961]	0.999 [0.996, 1.000]	0.08 [0.04, 0.10]
HELM	harmonic mean	77	0.906 [0.875, 0.958]	0.990 [0.968, 0.998]	0.08 [0.03, 0.11]
HELM	median	77	0.946 [0.920, 0.966]	0.991 [0.986, 0.995]	0.05 [0.03, 0.07]
HELM	min	77	0.838 [0.742, 0.902]	0.824 [0.719, 0.898]	-0.01 [-0.04, 0.00]
HELM	Borda	78	0.907 [0.878, 0.971]	0.962 [0.951, 0.987]	0.05 [0.01, 0.08]
HELM	Condorcet winner	78	0.740 [0.684, 0.946]	0.797 [0.751, 0.951]	0.06 [-0.00, 0.08]
HELM	plurality	77	0.722 [0.374, 0.862]	0.737 [0.299, 0.888]	0.02 [-0.16, 0.06]
OLLB	mean	4495	0.970 [0.968, 0.971]	1.000 [1.000, 1.000]	0.03 [0.03, 0.03]
OLLB	geometric mean	4495	0.940 [0.937, 0.944]	0.973 [0.970, 0.975]	0.03 [0.03, 0.04]
OLLB	harmonic mean	4495	0.899 [0.892, 0.905]	0.919 [0.911, 0.924]	0.02 [0.02, 0.02]
OLLB	median	4494	0.952 [0.937, 0.962]	0.971 [0.969, 0.973]	0.02 [0.01, 0.03]
OLLB	min	4494	0.848 [0.839, 0.857]	0.843 [0.834, 0.852]	-0.01 [-0.01, -0.00]
OLLB	Borda	101	0.909 [0.876, 0.936]	0.926 [0.893, 0.951]	0.02 [-0.00, 0.04]
OLLB	Condorcet winner	102	0.938 [0.916, 0.956]	0.930 [0.905, 0.953]	-0.01 [-0.02, 0.01]

Minimum versus mean by dispersion. Table 4 gives the median cost ratio $L_1^{\min}/L_1^{\text{mean}}$ in terciles of the target's coefficient of variation, with 95% bootstrap intervals over targets, at the ceiling and under the empirical caps (Open LLM Leaderboard; the four HELM targets that are feasible under the empirical caps are omitted).

Table 4: Median $L_1^{\min}/L_1^{\text{mean}}$ by tercile of the coefficient of variation (CV) of the target's task scores; "min cheaper" is the share of targets for which the minimum rule needs fewer points; Spearman correlation with CV over all targets.

dataset	cap	CV tercile	targets	CV range	$L_1^{\min}/L_1^{\text{mean}}$, median [95% CI]	min cheaper
HELM	ceil	low	26	0.10–0.17	0.083 [0.054, 0.100]	1.00
		mid	26	0.17–0.22	0.147 [0.118, 0.172]	1.00
		high	26	0.22–0.30	0.229 [0.217, 0.246]	1.00
		all, Spearman $+0.92$	78	0.10–0.30	0.147 [0.119, 0.179]	1.00
OLLB	ceil	low	1499	0.27–0.60	0.197 [0.193, 0.199]	1.00
		mid	1498	0.60–0.80	0.243 [0.242, 0.244]	1.00
		high	1498	0.80–1.84	0.352 [0.350, 0.354]	1.00
		all, Spearman $+0.84$	4495	0.27–1.84	0.251 [0.248, 0.253]	1.00
OLLB	emp. 95%	low	17	0.30–0.37	0.050 [0.041, 0.073]	1.00
		mid	16	0.37–0.42	0.069 [0.041, 0.095]	1.00
		high	16	0.42–0.52	0.136 [0.115, 0.189]	1.00
		all, Spearman $+0.66$	49	0.30–0.52	0.083 [0.063, 0.101]	1.00

Table 5: Minimum investment per rule, dataset and cap: feasible targets/targets, then median (Q1–Q3) tasks touched K and points L_1 over *feasible* targets only, so medians of rows with different feasible counts refer to different sets of targets and are not comparable. OLLB columns use the 102-target rank-stratified sample, so their medians are not population medians. Empirical 95% caps: per-task 95th percentile of observed Open LLM Leaderboard fine-tuning gains, applied to HELM through the MMLU-Pro value. The trimmed mean drops $t = 1$ of 6 scores at each end on OLLB and $t = 5$ of 57 on HELM (36 targets), so its two lines are not comparable.

	HELM MMLU (57 subjects)			OLLB sample (6 tasks)		
	feas./targets	K	L_1	feas./targets	K	L_1
cap 100 (ceil)						
mean	78/78	16 (10–23)	696 (443–1056)	102/102	1 (1–2)	56 (36–95)
median	78/78	21 (14–27)	112 (40–253)	102/102	2 (1–3)	21 (7–49)
min	78/78	12 (7–18)	98 (46–241)	102/102	2 (2–2)	16.8 (8.4–22.9)
trimmed mean	34/34 (+2 cens.)	11 (8–14)	327 (219–447)	102/102	1 (1–2)	45 (28–79)
Borda	78/78	47 (38–52)	534 (292–892)	102/102	3 (2–5)	23 (11–60)
Condorcet winner	78/78	29 (28–29)	213 (110–350)	102/102	4 (3–4)	35 (20–73)
plurality	78/78	9 (9–9)	47 (25–80)	102/102	2 (2–2)	26 (20–36)
cap +10 points						
mean	29/78	37 (23–46)	360 (222–453)	56/102	4 (4–5)	37 (32–48)
median	40/78	14 (10–18)	43 (18–70)	56/102	2 (1–3)	8.5 (3.3–16.5)
min	14/78	3 (2–5)	9.0 (2.6–17.8)	55/102	2 (2–2)	8.5 (6.3–12.0)
trimmed mean	28/36	31 (19–40)	292 (185–375)	41/102	3 (3–4)	27 (24–31)
Borda	25/78	40 (21–50)	185 (56–295)	65/102	2 (2–4)	12.4 (8.1–21.8)
Condorcet winner	38/78	28 (24–29)	107 (63–155)	40/102	4 (3–4)	18.3 (11.6–24.0)
plurality	57/78	9 (9–9)	31 (20–51)	13/102	2 (2–2)	10.4 (7.2–16.3)
cap emp. 95%						
mean	7/78	49 (11–51)	154 (32–162)	68/102	2 (2–4)	40 (32–55)
median	10/78	4 (2–9)	4.7 (0.7–15.9)	49/102	1 (1–2)	6.3 (3.1–16.1)
min	5/78	1 (1–2)	2.3 (2.0–2.4)	14/102	1 (1–2)	3.6 (2.9–6.3)
trimmed mean	7/36	39 (10–42)	119 (31–127)	41/102	2 (2–3)	27 (24–31)
Borda	8/78	28 (11–32)	50 (17–58)	59/102	2 (2–3)	11.9 (7.7–23.4)
Condorcet winner	10/78	18 (10–26)	23 (8–51)	16/102	3 (3–4)	11.3 (9.5–15.6)
plurality	16/78	8 (6–11)	12.3 (4.7–20.2)	39/102	2 (2–2)	23 (17–29)

Cap realism. Table 6 gives, per rule, the share of targets for which first place is infeasible under the per-task empirical 95% caps and the share whose h^* exceeds the pooled 95th percentile of fine-tuning gains (25.3 points). Ordinal rules on the Open LLM Leaderboard are over the 102-target rank-stratified sample in the all-target block (†), not over all 4495 targets.

Minimum cap by rank, task-subset scaling and robustness runs. Figure 4 shows the median h^* by rank, Figure 5 the task-subset scaling of Section 5.2 and Figure 6 the noise and task-resampling runs of Section 5.6.

Table 6: Share infeasible under the per-task empirical 95% cap (95% Wilson interval computed from the share and the number of targets; the interval is not degenerate when all targets agree, e.g. about [0, 0.16] for 0 of 20) and share of targets whose h^* exceeds the pooled empirical 95th percentile; all targets, then ranks 11–30. †: Open LLM Leaderboard ordinal rules, over the 102-target rank-stratified sample (weighted, over-representing top ranks; one Borda target unsolved), not comparable with the other rows of that block; in the ranks 11–30 block all rules use the same targets. HELM caps apply the MMLU-Pro per-task value (3.2 points) to every subject.

	HELM		OLLB	
	infeasible at emp. 95%	$h^* >$ pooled	infeasible at emp. 95%	$h^* >$ pooled
mean	0.91 [0.83, 0.96]	0.13	0.93 [0.92, 0.93]	0.69
geometric mean	0.94 [0.86, 0.97]	0.14	0.98 [0.97, 0.98]	0.63
harmonic mean	0.95 [0.88, 0.98]	0.19	0.98 [0.98, 0.99]	0.48
median	0.87 [0.78, 0.93]	0.12	0.97 [0.96, 0.97]	0.80
min	0.94 [0.86, 0.97]	0.38	0.99 [0.98, 0.99]	0.00
Borda	0.90 [0.81, 0.95]	0.22	0.42 [0.33, 0.52]†	0.16†
Condorcet winner	0.87 [0.78, 0.93]	0.12	0.84 [0.76, 0.90]†	0.19†
plurality	0.79 [0.69, 0.87]	0.03	0.62 [0.52, 0.71]†	0.16†
ranks 11–30				
mean	1.00 [0.84, 1.00]	0.00	0.00 [0.00, 0.16]	0.00
geometric mean	1.00 [0.84, 1.00]	0.00	0.30 [0.15, 0.52]	0.00
harmonic mean	1.00 [0.84, 1.00]	0.00	0.40 [0.22, 0.61]	0.00
median	0.95 [0.76, 0.99]	0.00	0.10 [0.03, 0.30]	0.00
min	1.00 [0.84, 1.00]	0.05	0.65 [0.43, 0.82]	0.00
Borda	1.00 [0.84, 1.00]	0.00	0.10 [0.03, 0.30]	0.00
Condorcet winner	0.95 [0.76, 0.99]	0.00	0.50 [0.30, 0.70]	0.00
plurality	0.65 [0.43, 0.82]	0.00	0.25 [0.11, 0.47]	0.00

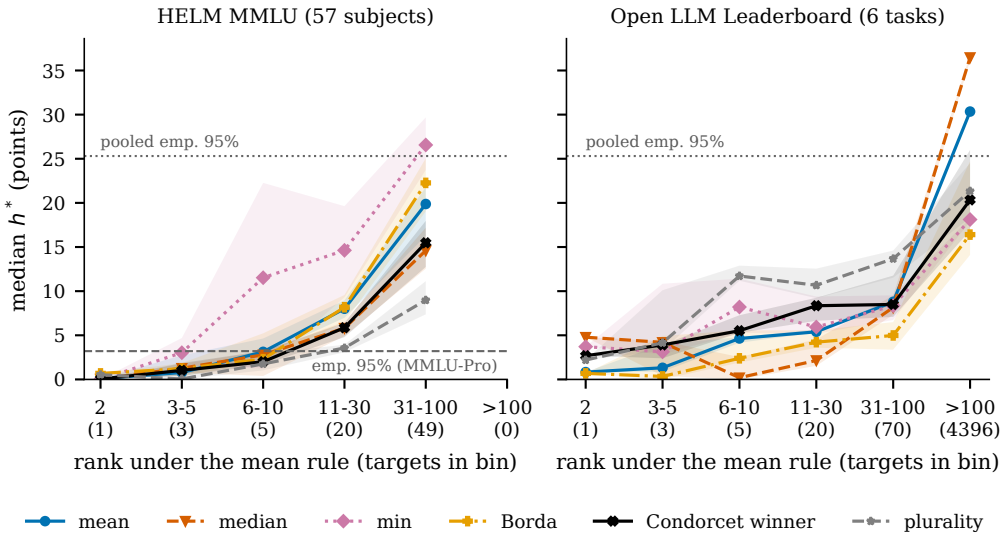


Figure 4: Median minimum uniform cap h^* (points) by the target's rank under the mean, per rule, with 95% bootstrap intervals over targets; targets per bin in brackets. Reference lines as in Figure 3.

Other caps. Table 7 reports the front+0, front+2, +5, +20 and empirical 99% caps in the layout of Table 5.

C Robustness details

Margin ε. A margin of 10^{-6} instead of 10^{-3} changes no feasibility and no median cost (largest per-group ratio 1.000 on HELM and 1.000 on the Open LLM Leaderboard). A margin of 0.1 raises the median L_1 by factors 1.008–1.033 (HELM) and 1.003–1.019 (Open LLM Leaderboard) across rules at the ceiling and cap +10, and turns at most 1 HELM and 10 Open LLM Leaderboard targets from feasible to infeasible in any (rule, cap) cell; single targets whose base cost is small can change by much larger factors.

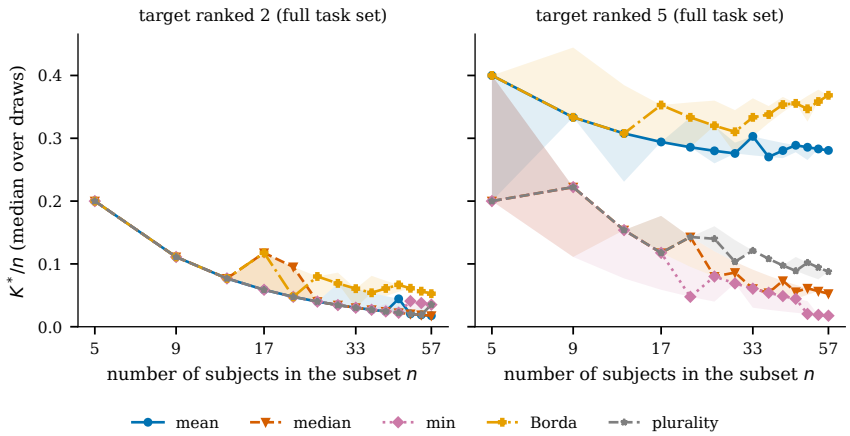


Figure 5: K/n for random subsets of n of the 57 HELM subjects (50 draws per n, cap $+10$), for the models ranked 2 (left) and 5 (right). Line: median over feasible draws; band: bootstrap interval of the median.

Slices of the Open LLM Leaderboard. Table 8 compares the full leaderboard (ceiling cap, rank-stratified sample: 20 targets of rank 11–30, 70 of rank 31–100) with the merge-free slice, the raw (un-normalised) scores and the top-500 slice. Dropping merged models moves most medians by a few points at most (ranks 11–30: from -1.7 for the minimum to $+2.1$ for the Condorcet requirement, $+0.9$ for the mean; ranks 31–100: up to $+3.5$ for the median, -2.8 for the minimum) but the order of the median and the minimum rule is not robust to the slice (next paragraph). For the score rules the top-500 slice equals the full leaderboard (a target's cost depends only on the models above it), and only the ordinal rules differ, by up to 3 points. The raw scale changes the picture: the median rule's cost in ranks 11–30 rises from 4.2 to 20.8 points while the mean's falls from 32.2 to 23.0, so the normalisation matters for rules that use the order or the spread of task scores. Intervals are bootstrap intervals over the sampled targets; they do not cover the choice of slice. Cap $+10$ runs exist for all three slices (summarised in the released slice file) but are not tabulated here.

Median versus minimum across slices. The slice medians use different target sets: over the sampled targets of rank 2–100 (ceiling cap; 54 targets in the main sample and the merge-free and raw slices, 99 in the top-500 slice) the median L_1 of the median rule is 7.7 and that of the minimum rule 10.7 in the main analysis, so the median rule is cheaper; without merged models the two are 14.0 and 9.4, on raw scores 22.2 and 13.2, and in the top-500 slice 15.3 and 12.6, so the minimum is cheaper in all three. A statement that one of these two rules is cheaper on the Open LLM Leaderboard is therefore conditional on the data slice (the mean rule costs 37.0, 37.8, 28.1 and 48.2 in the four slices).

Spillover (stretch). If effort on one task lowers the other tasks by $\kappa(D-d_t)$ with $\kappa = \lambda/(n-1)$, the exact solvers give a median L_1 ratio to the no-spillover cost of 1.11 ($\lambda = 0.1$) and 1.33 ($\lambda = 0.25$) for the mean, 1.00 and 1.01 for the minimum on HELM, and 1.02, 1.05 on the Open LLM Leaderboard; for the mean the ratio equals $1/(1-\lambda)$. Spillover therefore raises costs most for the mean and least for the minimum. The solver at $\lambda = 0$ reproduces the main solvers.

D Further sensitivity analyses

These analyses repeat parts of the main grids under changed assumptions; none changes the headline comparison of the minimum with the mean, and two change the rule order under gain caps. Targets are those of the main grids (HELM: 78 non-leaders; Open LLM Leaderboard: ranks 2–100, 99 targets, ceiling cap, margin $\varepsilon = 10^{-3}$ unless stated).

Convex task cost. The main grids charge one unit per score point. We re-solved the minimum-cost problem exactly for the mean and the minimum under a quadratic cost $(y_t - x_t)^2$ (a large jump is dearer) and a log-headroom cost $\ln \frac{100-x_t}{100-y_t}$ (points near the ceiling are dearer); the linear column

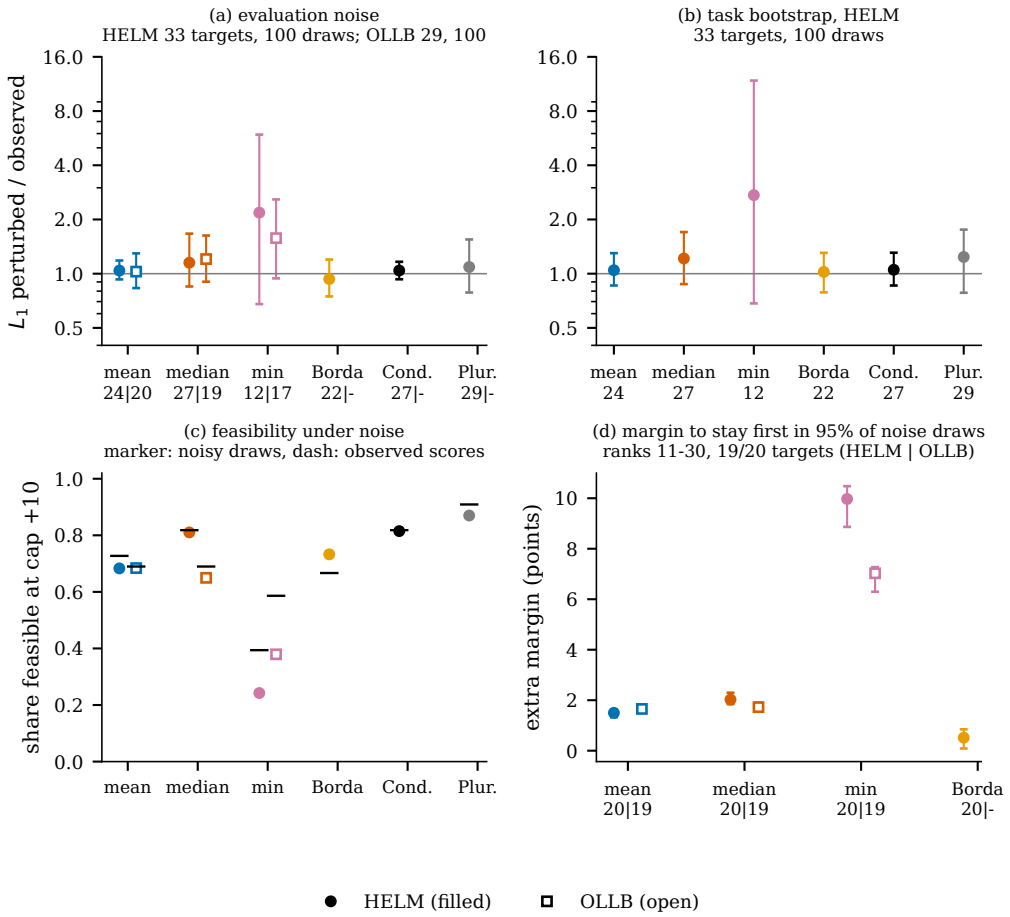


Figure 6: Noise and task resampling on rank-stratified samples (cap +10; HELM filled, Open LLM Leaderboard open; numbers of targets and draws are in the panel titles, HELM evaluation noise has 100 draws). Bars are 95% intervals across draws; the number of targets entering each rule's ratio (HELM|Open LLM Leaderboard; "-": not run) is under the rule name. (a) Ratio of L_1 under evaluation noise to L_1 at the observed scores. (b) The same ratio under a bootstrap over HELM subjects. (c) Share of target-draws that are feasible under noise (marker) against the observed scores (dash). (d) Extra raise beyond ε for a just-first target to stay first in 95% of noise draws (ranks 11–30). The Condorcet and plurality rules were not run for (d), and the ordinal rules were not run on the Open LLM Leaderboard here, hence the missing symbols.

reproduces the main-grid costs. Median ratio $L_1^{\min}/L_1^{\text{mean}}$ (linear / quadratic / log-headroom): HELM 0.15 / 0.17 / 0.09, Open LLM Leaderboard 0.28 / 0.27 / 0.25. The minimum is cheaper than the mean for 100% / 97% / 100% of HELM targets and 100% / 88% / 100% of Open LLM Leaderboard targets. The finding therefore holds for most targets but is not universal under a quadratic cost: on the Open LLM Leaderboard the mean is cheaper for 27% of the high-dispersion tercile (median ratio there 0.45). That the ratio rises with dispersion persists (Spearman correlation with the coefficient of variation: HELM 0.92 linear, 0.52 quadratic; Open LLM Leaderboard 0.72 and 0.68). Costs are stylised and equal across tasks; this tests the shape of the cost, not task-specific prices.

Noise-robust margins. Because $\varepsilon = 10^{-3}$ is not a robust margin (Section 5.6), we re-ran the comparison at margins 0.1 and 1 point and at rule-specific margins (2 points for the mean and the median, 8 for the minimum, about the extra margin each needs for a 95% stay probability). The minimum remains cheaper than the mean for every target in all four settings; its median cost ratio is 0.15 (base), 0.15 (0.1), 0.15 (1) and 0.27 (rule-specific) on HELM and 0.28, 0.28, 0.29 and 0.50 on the Open LLM Leaderboard, so the advantage roughly halves with rule-specific margins, and the correlation with dispersion stays high (0.95 and 0.62). The rule order under fixed gains changes: at

Table 7: Minimum investment under the remaining caps, same layout as Table 5. The trimmed mean was not run under these caps (n/a).

	HELM MMLU (57 subjects)			OLLB sample (6 tasks)		
	feas./targets	K	L_1	feas./targets	K	L_1
cap front+0						
mean	78/78	37 (29–43)	696 (443–1056)	102/102	3 (2–3)	56 (36–95)
median	78/78	21 (14–27)	116 (44–271)	102/102	2 (1–3)	21 (7–49)
min	78/78	12 (7–18)	98 (46–241)	102/102	2 (2–2)	16.8 (8.4–22.9)
trimmed mean	34/34 (+2 cens.)	28 (21–32)	327 (219–447)	102/102	2 (2–3)	45 (28–79)
Borda	78/78	48 (40–52)	549 (308–909)	102/102	3 (2–5)	23 (11–60)
Condorcet winner	78/78	31 (29–32)	259 (130–431)	102/102	4 (3–4)	35 (20–73)
plurality	78/78	18 (18–18)	149 (85–237)	102/102	3 (3–3)	45 (37–59)
cap front+2						
mean	78/78	31 (23–38)	696 (443–1056)	102/102	2 (2–3)	56 (36–95)
median	78/78	21 (14–27)	112 (40–253)	102/102	2 (1–3)	21 (7–49)
min	78/78	12 (7–18)	98 (46–241)	102/102	2 (2–2)	16.8 (8.4–22.9)
trimmed mean	n/a	n/a	n/a	n/a	n/a	n/a
Borda	78/78	47 (38–52)	534 (292–892)	102/102	3 (2–5)	23 (11–60)
Condorcet winner	78/78	29 (28–29)	213 (110–350)	102/102	4 (3–4)	35 (20–73)
plurality	78/78	9 (9–9)	47 (25–80)	102/102	2 (2–2)	26 (20–36)
cap +5						
mean	11/78	34 (20–49)	168 (98–241)	13/102	6 (5–6)	28 (24–29)
median	16/78	9 (4–10)	16.6 (2.8–21.0)	31/102	1 (1–2)	3.7 (2.3–5.1)
min	7/78	2 (1–2)	2.4 (2.2–5.3)	16/102	1 (1–2)	3.8 (3.0–6.7)
trimmed mean	n/a	n/a	n/a	n/a	n/a	n/a
Borda	11/78	24 (16–40)	56 (32–127)	33/102	3 (2–4)	8.8 (5.6–12.4)
Condorcet winner	16/78	24 (16–27)	57 (20–74)	6/102	3 (2–4)	7.6 (6.4–9.4)
plurality	29/78	9 (8–10)	21 (10–26)	3/102	2 (2–2)	5.5 (4.5–6.5)
cap +20						
mean	56/78	28 (19–39)	541 (356–754)	81/102	3 (2–4)	47 (33–64)
median	66/78	19 (12–25)	80 (37–182)	78/102	2 (1–2)	15.6 (4.7–22.4)
min	34/78	6 (3–9)	44 (14–55)	95/102	2 (2–2)	14.8 (8.1–21.7)
trimmed mean	n/a	n/a	n/a	n/a	n/a	n/a
Borda	49/78	44 (34–49)	331 (184–485)	80/102	2 (2–4)	14.7 (8.9–30.7)
Condorcet winner	65/78	29 (28–29)	169 (103–272)	78/102	4 (3–4)	24 (17–41)
plurality	74/78	9 (9–9)	46 (25–74)	72/102	2 (2–2)	23 (18–27)
cap emp. 99%						
mean	15/78	32 (22–42)	222 (155–291)	74/102	2 (2–3)	44 (32–59)
median	26/78	10 (8–14)	27 (14–38)	70/102	2 (1–2)	14.0 (4.1–21.9)
min	9/78	2 (1–3)	3.0 (2.3–7.6)	36/102	2 (1–2)	6.7 (4.5–9.2)
trimmed mean	n/a	n/a	n/a	n/a	n/a	n/a
Borda	16/78	33 (20–41)	120 (51–173)	75/102	2 (2–4)	13.6 (8.8–27.3)
Condorcet winner	25/78	27 (18–28)	77 (29–106)	48/102	4 (3–4)	20 (14–28)
plurality	45/78	9 (9–10)	27 (19–38)	54/102	2 (2–2)	26 (17–36)

cap $+10$ the share of targets for which first is feasible is, under the base margin and rule-specific margins, 37% and 19% (mean), 51% and 41% (median), 18% and 4% (minimum) on HELM, and 100% and 49%, 76% and 61%, 69% and 6% on the Open LLM Leaderboard. The minimum, which needs the largest margin, is hardly ever feasible under rule-specific margins, and on the Open LLM Leaderboard the median overtakes the mean. Statements about which rule is most or least feasible under fixed gains are thus conditional on the margin.

Dependence between targets: clusters by base model. The Open LLM Leaderboard targets include fine-tunes and merges of one another. We resampled whole clusters defined by the user-entered `Base Model` field (3869 clusters for 4495 targets; models without a usable entry form their own cluster; 1000 resamples). The median cost ratio of the minimum to the mean is 0.251 with a cluster interval [0.247, 0.256] against a target-level interval [0.248, 0.253], and 0.352 [0.342, 0.355] in the high-dispersion tercile. The difference of R^2 of $\log L_1$ on $\log$ gap and on $\log$ rank, with cluster intervals, is 0.032 [0.029, 0.036] (geometric mean), 0.020 [0.017, 0.023] (harmonic), 0.020 [0.010, 0.035] (median) and -0.005 [-0.008, -0.002] (minimum); for Borda and the Condorcet requirement (ordinal sample) the intervals include zero. Clusters widen the intervals only slightly and change no

Table 8: Open LLM Leaderboard slices, rank bin 11–30, ceiling cap: median K and L_1 by rule.

rule	full	no merged	raw scores	top 500
ranks 11–30: median L_1 [95% CI] (K)				
mean	32.2 [31.7, 33.9] (1)	33.1 [31.9, 34.8] (1)	23.0 [20.5, 24.0] (1)	32.2 [31.8, 33.9] (1)
median	4.2 [3.0, 10.0] (1)	4.8 [3.2, 12.8] (1)	20.8 [17.1, 22.5] (2)	4.2 [3.0, 10.0] (1)
min	9.7 [6.7, 17.6] (2)	8.0 [5.3, 15.5] (2)	10.1 [7.3, 13.5] (2)	9.7 [6.7, 17.6] (2)
Borda	9.4 [7.7, 12.1] (2)	9.1 [6.7, 12.2] (1)	8.3 [5.1, 9.5] (2)	6.7 [4.2, 11.9] (1)
Condorcet winner	17.6 [12.8, 20.6] (3)	15.5 [12.0, 20.2] (3)	14.8 [12.1, 16.8] (3)	17.6 [12.8, 20.6] (3)
plurality	17.5 [16.3, 19.9] (2)	17.1 [13.5, 18.6] (2)	22.5 [19.4, 23.3] (2)	14.6 [9.8, 15.7] (2)
ranks 31–100: median L_1 [95% CI] (K)				
mean	52.6 [50.8, 53.8] (1)	52.0 [47.5, 55.0] (1)	36.1 [32.3, 38.7] (1)	52.6 [50.9, 53.7] (1)
median	17.9 [15.7, 21.0] (2)	21.4 [17.0, 22.1] (2)	30.8 [26.4, 33.0] (2)	17.9 [15.9, 21.0] (2)
min	13.8 [11.3, 16.4] (2)	11.0 [6.8, 15.7] (2)	14.3 [13.4, 15.5] (2)	13.8 [11.2, 16.1] (2)
Borda	14.5 [11.9, 20.2] (2)	18.5 [12.4, 23.9] (3)	14.8 [11.4, 16.7] (2)	19.9 [19.0, 21.5] (3)
Condorcet winner	26.3 [23.1, 31.9] (4)	28.5 [22.2, 32.8] (4)	21.4 [19.1, 24.9] (4)	31.9 [29.4, 33.0] (4)
plurality	22.5 [21.5, 25.6] (2)	19.9 [17.9, 21.1] (2)	22.6 [19.6, 24.5] (2)	19.2 [17.7, 21.8] (2)

conclusion. The field is user-entered: 2612 of the 540 fine-tune/base pairs of the empirical gains have a base entry that is a placeholder string or does not resolve, so some clusters are incomplete, and 19 pairs have the fine-tune below its base on all six tasks. HELM has no such field, so its intervals stay at the level of targets.

Composition of the empirical-gain pairs. Of the 540 pairs, 346 (64%) are fine-tuned on domain-specific data and 154 (29%) are chat, RLHF or preference-tuned models, which are trained for several tasks at once and may not be informative about a developer who targets one task. Dropping chat-typed models, merges and fine-tunes whose name signals instruction tuning leaves 239 pairs; the pooled 95% quantile falls from 25.3 to 19.0 points, and the shares of targets with first place infeasible under the per-task caps barely change (ranks 11–30 on HELM: mean 100% to 100%, median 95% to 95%, minimum 100% to 100%; ranks 2–30 on the Open LLM Leaderboard: mean 0% to 0%, median 7% to 7%, minimum 62% to 72%).

Ordinal rules for every target of ranks 2–100. In the main text the Open LLM Leaderboard ordinal rules use a stratified sample; here we ran Borda, plurality and the Condorcet requirement for all 99 targets of ranks 2–100. At the ceiling every target is feasible for each of them; the median number of tasks touched and the median L_1 are Borda 2 tasks and 17.7 points, plurality 2 and 22.1, Condorcet 3 and 26.2, against 1 and 48.2 for the mean and 2 and 12.6 for the minimum. At cap $+10$ the share with first place feasible is Borda 93%, Condorcet 54%, plurality 14% and, for the score rules, mean 100%, geometric mean 83%, median 76%, harmonic mean 72%, minimum 69%; plurality is the hardest rule to make feasible under a fixed gain on this dataset, as in the sample.

Computation. The released jobs used about 24 CPU-hours as metered by the scheduler (summed over the jobs finished at the time of the last snapshot; the host was shared), about five times the 4 CPU-hour estimate in the task description. The largest solve for the ordinal rules peaked at 669 MB resident memory.

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